AMIT MISHRA

GCP Data & Infrastructure Engineer | Cloud Architect | AI-Enhanced Data Solutions

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Summary

Experienced GCP Data and Infrastructure Engineer with 10 years of expertise in designing, migrating, and optimizing cloud-native data platforms. Skilled in Terraform-driven infrastructure automation, cloud migration strategies including Teradata and Hive to BigQuery/GCP, and multi-region to single-region cloud consolidations. Proficient in leveraging AI-driven developer tools such as **Cursor AI** and **CodeRabbit** to enhance productivity in code generation, review, and automation workflows. Experienced integrating **TensorFlow** AI components and **Gemini AI** capabilities for intelligent data orchestration and automation within GCP environments.

Proven expertise in performance optimization and cloud cost management, achieving cost savings of **$2.5M+** across GCP.

Skills

Google Cloud Platform: BigQuery (data modeling & optimization), Dataflow, Dataproc, Pub/Sub, Cloud Storage, Cloud Composer (Airflow), Vertex AI, AI Platform (AI integration), GKE, App Engine, Cloud SQL, Compute Engine, Cloud Functions, Cloud Run, Cloud Bigtable, Cloud AutoML, Cloud IAM, Cloud Monitoring, BigQuery ML, Gemini Generative AI

Cloud Infrastructure & Automation: Terraform, Kubernetes, Multi-region to Single-region Consolidation

Data Modeling: Star Schema, Snowflake Schema, Normalization, Denormalization, Dimensional Modeling

Migration Expertise: Teradata to GCP, Hive to BigQuery, ETL/ELT Modernization

BigQuery Optimization: Partitioning, Clustering, Slot Management, Materialized Views, Cost Controls

Developer AI Tools: Cursor AI (code generation & review), CodeRabbit (automation & testing)

AI Integration: TensorFlow for AI components, Gemini AI for intelligent workflow automation

Programming & Orchestration: Python, SQL, Shell Scripting, Airflow, Workflow Automation

Version Control & CI/CD: Git, GitHub Actions, GitLab CI

Monitoring & Security: Stackdriver Monitoring, IAM, VPC, Data Encryption, Cloud Security Best Practices

- Terraform GitHub Networking BigQuery Bigtable DataflowCloud Storage and other GCP Cloud Technology Services
- Experience in managing Google Compute EngineGCE BigQuery Cloud Composer Google Kubernetes EngineGKE Cloud Storage and other Google Cloud Managed Service
- Experience with Monitoring tool like Stackdriver Monitoring Logging Debugging and Tracing on Google Cloud Platform and knowledge on creating metrics for custom logs
- Experience with monitoring and log analytical tools like GrafanaLogscale
- Experience in Cloud computing concepts like Virtual Private Cloud VPC DNS CDN Load Balancing Auto Scaling BigQuery Bitable and Dataflow
- Supporting Monolithic and Microservices workload along with integration part with CICD tools
- Experience in managing build and deployment using Terraform CloudJenkins and GitHub
- Strong experience in managing Linux systems
- Managing backups and patching across the environments
- Diagnose and resolve customers complex technical issues for Google Cloud Platform products
- Resolution of critical and complex technical issues that will impact customer business and Service levels
- Provide ongoing support for Google Cloud infrastructure
- Prepare and execute changes as per change management process
- Prepare RCA for the Sev 1 incidents and drive the corrective action plans if any based on the RCA
- Initiate the problem management process for repeated incidents and drive for permanent fix
- Carry out performance finetuning activities

- Automate daily activities to improve the Operational Governance
- Deviserefineimplement and drive Cloud operations best practices
- Coordinate and work with internal teams to for Incident Change and problem tickets
- Prepare SOPs for the regular activities and update KEDB
- Regular update of architecture and technical documentation

Experience

Groupon India – Senior GCP Data Engineer

Feb 2021 – Present

- Designed and modernized Teradata and Hive data warehouses into optimized BigQuery/GCP environments with robust data modeling for analytics and reporting.
- Led migration from multi-region to single-region cloud architecture with automated Terraform provisioning and Git-based CI/CD.
- Developed an AI agent to reduce migration time by 8 to 10 hours for each pipeline, significantly accelerating project delivery timelines.
- Created a Data Transfer tool that reduced data transfer time by 99%, dramatically improving operational efficiency.
- Reduced 37% of cost to bring down data platform cost under $5 Million through strategic optimization and resource management.
- Actively engaged with developer teams to reduce incident and ticket volume by 50%, improving overall system stability.
- Automated repetitive tasks by introducing self-healing data pipelines, reducing KTLO (Keep The Lights On) by 20 hours per week.
- Employed Cursor AI and CodeRabbit to accelerate code reviews, automate test case generation, and streamline data pipeline development.
- Integrated TensorFlow-driven AI components for intelligent data processing tasks and employed Gemini AI to automate repetitive cloud workflows.
- Implemented cost control mechanisms including slot rescaling, query optimization, and data partitioning to reduce cloud spend significantly.
- Led a team of 15 professionals in the end-to-end migration of legacy data platforms (Teradata and Hive) to a modern, scalable cloud data warehouse (BigQuery and Snowflake) for a major US Telecom client.
- Architected and implemented BigQuery performance optimization strategies, including Partitioning, Clustering, and Reservations, resulting in an estimated annual cost savings of $1.5 Million.

- Designed and built scalable ETL/ELT data pipelines utilizing Google Cloud Dataflow (Apache Beam) and Dataform for transformations, ensuring data governance and quality in the new platform.
- Spearheaded the integration of BigQuery with tools like Data Studio/Looker/Grafana, and developed custom solutions for cost tracking and lineage mapping, contributing to GCP Cost Optimization.
- Migrated buckets from multi-region to single-region Cloud.
- Integrated streaming pipelines.
- Worked on SOX enablement in BigQuery.
- Enabled column security using masking.
- Performed Dataproc cluster resizing and optimization.
- Enabled Autoclass, lifecycle management, and storage class optimizations.

IBM India – GCP Data Engineer

Sept 2019 – Feb 2021

- Designed and architected scalable, secure, and highly available cloud infrastructure using GCP services like Compute Engine, GKE, Cloud Storage, and VPC networking.
- Automated infrastructure provisioning and configuration using Infrastructure as Code (IaC) tools such as Terraform and Deployment Manager to ensure consistency across environments.
- Monitored cloud resources and applications to optimize performance, troubleshoot issues, and manage resource utilization efficiently.
- Implemented security best practices, including IAM, data encryption, firewall rules, and regular security audits to protect cloud assets.
- Managed disaster recovery strategies and data backup plans leveraging GCP's redundancy features for business continuity.
- Designed and managed complex network configurations, including VPNs, subnets, load balancers, and Cloud CDN for reliable and optimized connectivity.
- Collaborated with DevOps and development teams to build and maintain CI/CD pipelines, supporting rapid and reliable software deployments.
- Analyzed and optimized cloud costs through resource rightsizing, autoscaling, and budget monitoring.
- Worked cross-functionally with stakeholders, including security, operations, and development teams, to translate business requirements into technical cloud solutions.

Teradata Corporation – Data Engineer

Feb 2018 – Sept 2019

- Supported large-scale Teradata and Hive to BigQuery migration projects focusing on data transformation, infrastructure automation with Terraform, and version control with Git.
- Conducted cost optimization efforts and infrastructure tuning.

Tech Mahindra – Junior Data Engineer

Apr 2015 – Feb 2018

- Maintained legacy Teradata Datawarehouse systems with performance tuning and optimization.

Education

B.Tech in Electronics and Communication Engineering
RGTU, Bhopal, 2014

Certifications

- Google Cloud Professional Data Engineer (expected/completed)
- GCP Associate Cloud Engineer
- Terraform Associate Certification (if applicable)
- Teradata Database Associate 16.10
- Teradata DBA Certified